

An Inventory-Driven Dynamical Vacuum Cosmology: Higher H_0 , Lower S_8 , and Marginally Improved Fits Relative to Matched Λ CDM

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Abstract

We investigate a phenomenological cosmological framework in which the effective vacuum energy density evolves according to the growth of a closure inventory rather than a fundamental scalar field.

The model is based on a single inventory principle,

$$\rho_\Lambda(a) \propto I(a)^{-1},$$

where $I(a)$ denotes the cumulative closure inventory of the universe.

The inventory evolves through the formation, stabilization, and accumulation of effective closure structures.

Unlike conventional early-dark-energy scenarios, the present framework introduces no inflaton potential and no additional scalar degrees of freedom.

A native implementation in the CLASS Boltzmann code is developed and tested against the combination of Planck 2018 CMB, BAO, and Pantheon+ supernova data.

Preliminary MCMC analyses indicate that the inventory cosmology provides fits comparable to, and marginally better than, matched Λ CDM runs while naturally shifting the inferred Hubble constant toward larger values.

The present implementation is restricted to post-inflationary cosmological evolution.

1 Introduction

The standard cosmological model (Λ CDM) successfully explains a broad range of cosmological observations.

Nevertheless, persistent discrepancies remain, most notably the Hubble tension.

A wide variety of dynamical dark-energy and early-dark-energy models have been proposed. Most introduce additional scalar fields, modified potentials, or extra sectors.

In this work we explore a different possibility.

We assume that the effective vacuum energy density is not fundamental but instead depends on the cumulative amount of realized closure inventory in the universe.

The central hypothesis is

$$\rho_\Lambda \propto \frac{1}{I}.$$

The inventory is assumed to be non-decreasing,

$$I'(a) \geq 0,$$

which guarantees a monotonic depletion of the effective vacuum reservoir.

The present work is restricted to post-inflationary cosmology.

We do not attempt to replace primordial inflation in the current implementation.

Instead, we investigate whether the inventory principle alone can provide a viable late-time cosmological framework capable of alleviating the Hubble tension. The present analysis should be interpreted as a post-inflationary effective cosmology. We do not model the primordial inflationary phase, the generation of primordial perturbations, or reheating in this work. Instead, the goal of the present paper is more limited: to test whether the closure-inventory vacuum principle can provide a viable late-time cosmological framework and a CMB-compatible effective realization after the primordial initial conditions have already been established. Therefore, the present implementation assumes that primordial initial conditions are already given and investigates only the subsequent evolution of the universe driven by the inventory mechanism.

2 Inventory Principle

The total inventory is assumed to be

$$I(a) = I_{\text{vac}} + N_h(a)A_p(T) + N_s(a)A_s.$$

where

- I_{vac} is the irreducible vacuum inventory floor,
- N_h is the accumulated closure-history inventory,
- N_s denotes stable closure remnants,
- $A_p(T)$ is the temperature-dependent closure area,
- A_s is the stable closure area.

The temperature-dependent closure radius is

$$r_p(T) = \frac{r_{p0}}{1 + (T/T_{\text{QCD}})^\beta}.$$

The effective closure area becomes

$$A_p(T) = C_A 4\pi r_p(T)^2.$$

The vacuum density is postulated to obey

$$\rho_\Lambda(a) = \rho_{\Lambda,\text{ref}} \frac{I_{\text{ref}}}{I(a)}.$$

3 Closure Formation Dynamics

The creation window is modeled as

$$W(T) = \exp \left[-\frac{(\ln(T/T_Q))^2}{2\sigma_Q^2} \right].$$

A smooth freeze-out function is introduced:

$$F(T) = \frac{1}{1 + \exp[-\ln(T/T_f)/\sigma_f]}.$$

The pair-production rate is

$$G_p = \gamma_p W(T) F(T) \left(\frac{\rho_\Lambda}{\rho_{\Lambda, \text{ref}}} \right) \left(1 - \frac{N_p}{N_{p, \text{max}}} \right).$$

The populations evolve according to

$$\frac{dN_p}{d \ln a} = G_p - \lambda_{\text{ann}} N_p^2 - \lambda_s N_p,$$

$$\frac{dN_h}{d \ln a} = G_p \left(1 - \frac{N_h}{N_{h, \text{max}}} \right),$$

$$\frac{dN_s}{d \ln a} = \epsilon_B \lambda_s N_p \left(1 - \frac{N_s}{N_{s, \text{max}}} \right).$$

Inventory growth is

$$\frac{dI}{d \ln a} = A_p \frac{dN_h}{d \ln a} + A_s \frac{dN_s}{d \ln a} + N_h \frac{dA_p}{d \ln a}.$$

4 Energy Transfer and Conservation

The inventory principle implies

$$\rho'_\Lambda = -Q,$$

with

$$Q = \rho_\Lambda \frac{I'}{I}.$$

The vacuum energy released through inventory growth is phenomenologically distributed into radiation and matter sectors:

$$Q = Q_R + Q_M.$$

The continuity equations become

$$\rho'_R + 4\rho_R = Q_R,$$

$$\rho'_M + 3\rho_M = Q_M.$$

Therefore total energy conservation is preserved:

$$\rho'_\Lambda + \rho'_R + \rho'_M + 4\rho_R + 3\rho_M = 0.$$

In the present implementation, the vacuum transfer is phenomenologically partitioned between radiation and matter sectors while preserving total energy conservation.

5 Vacuum Depletion and Effective Equation of State

Differentiating

$$\rho_\Lambda \propto I^{-1}$$

yields

$$Q = \rho_\Lambda \frac{I'}{I}.$$

The effective equation of state becomes

$$w(a) = -1 + \frac{Q}{3\rho_\Lambda}.$$

Thus, when

$$I'(a) \rightarrow 0,$$

the model naturally approaches

$$w \rightarrow -1.$$

6 Implementation in CLASS

The model was implemented natively in CLASS v3.3.4.

At the background level, the following V9 ingredients are implemented explicitly:

- the closure inventory

$$I(a) = I_{\text{vac}} + N_h(a)A_p(T) + N_s(a)A_s,$$

- the inverse-inventory vacuum scaling

$$\rho_\Lambda(a) = \rho_{\Lambda,\text{ref}} \frac{I_{\text{ref}}}{I(a)},$$

- the monotonic inventory-growth condition

$$I'(a) \geq 0,$$

- the vacuum-depletion rate

$$Q = \rho_\Lambda \frac{I'}{I},$$

- the effective dark-energy equation of state

$$w(a) = -1 + \frac{Q}{3\rho_\Lambda},$$

- and the phenomenological energy transfer into radiation and matter sectors,

$$Q = Q_R + Q_M.$$

Thus, the background implementation realizes the core inventory mechanism directly: closure formation changes $I(a)$, the inventory determines $\rho_\Lambda(a)$, and the vacuum depletion determines $Q(a)$ and $w(a)$.

At the perturbation level, the present implementation is more limited. Perturbations are treated through an effective fluid closure approximation. That is, the V9 vacuum sector is represented as an effective dark-energy fluid in the perturbation equations rather than through a fully gauge-invariant perturbation theory of the microscopic inventory variables N_p , N_h , and N_s .

Therefore, the following components are not yet implemented in the present version:

- a fully gauge-invariant inventory perturbation formalism,
- independent perturbation variables for δN_p , δN_h , and δN_s ,
- a microphysical derivation of the radiation–matter transfer split Q_R/Q_M ,
- primordial inflationary dynamics,
- primordial perturbation generation,
- reheating or pre-inflationary inventory evolution.

The present work should therefore be understood as a post-inflationary effective realization of the V9 closure-inventory mechanism. A complete perturbative and primordial implementation is left for future work.

7 Data and Methodology

We compare the model against

- Planck 2018 TTTEEE likelihoods,
- BAO measurements,
- Pantheon+ Type Ia supernovae.

Parameter estimation is performed using Cobaya MCMC.

Matched Λ CDM runs are performed using identical likelihood combinations for direct comparison. For a fair comparison, the matched Λ CDM and V9 analyses were performed using identical likelihood combinations and the same number of cosmological sampling parameters. Therefore, direct comparisons of the best-fit χ^2 values are meaningful within the present analysis. A complete model-selection analysis including information criteria such as AIC and BIC is left for future work.

8 Results

The inventory cosmology provides a marginal improvement over matched Λ CDM while simultaneously increasing H_0 and lowering S_8 .

Parameter	Λ CDM	V9
Best-fit χ^2	4142.99	4140.92
H_0 [km s $^{-1}$ Mpc $^{-1}$]	$67.94^{+0.44}_{-0.46}$	$71.36^{+0.47}_{-0.46}$
$\Omega_b h^2$	$0.02237^{+0.00014}_{-0.00014}$	$0.02237^{+0.00013}_{-0.00014}$
$\Omega_c h^2$	$0.11979^{+0.00102}_{-0.00098}$	$0.11981^{+0.00098}_{-0.00100}$
n_s	$0.9652^{+0.0037}_{-0.0037}$	$0.9651^{+0.0038}_{-0.0037}$
τ_{reio}	$0.0539^{+0.0075}_{-0.0072}$	$0.0541^{+0.0074}_{-0.0074}$
A_{Planck}	$1.0006^{+0.0025}_{-0.0025}$	$1.0007^{+0.0025}_{-0.0025}$
Ω_m	$0.308^{+0.006}_{-0.006}$	$0.279^{+0.005}_{-0.005}$
σ_8	0.811	0.820
S_8	0.822	0.791
Gelman–Rubin $R - 1$	0.020	0.019

Table 1: Comparison between matched Λ CDM and V9 MCMC analyses using identical likelihood combinations (Planck 2018 + BAO + Pantheon+). The V9 model yields a higher inferred Hubble constant, a lower S_8 , and a marginally improved best-fit χ^2 relative to matched Λ CDM.

9 Discussion

The present framework should be interpreted primarily as a phenomenological post-inflationary realization of the closure-inventory principle. The most distinctive feature of the framework is that the effective dark-energy equation of state is not imposed phenomenologically.

Instead, the sequence

$$I(a) \rightarrow \rho_\Lambda(a) \rightarrow Q(a) \rightarrow w(a)$$

emerges automatically from the inventory principle

$$\rho_\Lambda \propto I^{-1}.$$

In contrast to conventional dynamical dark-energy models, the effective equation of state is not postulated but emerges directly from the inventory dynamics.

Future work will investigate fully gauge-invariant inventory perturbations, primordial dynamics, and strong-gravity applications. The numerical results should therefore be interpreted with appropriate caution. The present implementation tests whether the inventory principle can provide a viable post-inflationary cosmological background and effective perturbative realization. It does not yet constitute a complete replacement for inflation, nor does it provide a first-principles microscopic derivation of all closure variables. Nevertheless, the fact that the same fixed implementation yields a higher H_0 , lower S_8 , and a marginally improved best-fit χ^2 relative to matched Λ CDM motivates further investigation.

10 Future Work

Several extensions are required before the present framework can be regarded as a complete cosmological theory.

First, a full gauge-invariant perturbation implementation should be developed. The current analysis implements the V9 inventory mechanism at the background level and treats perturbations through an effective fluid closure approximation. A future version should introduce explicit perturbations of the inventory variables,

$$\delta N_p, \quad \delta N_h, \quad \delta N_s,$$

and construct gauge-invariant combinations suitable for direct use in the Einstein–Boltzmann system.

Second, the phenomenological transfer split

$$Q = Q_R + Q_M$$

should be replaced by a microphysical prescription. In the present work, the split between radiation and matter is treated effectively in order to preserve total energy conservation. A more fundamental treatment should derive the transfer fractions from the closure-formation dynamics.

Third, the model should be tested with additional likelihood combinations. In particular, future MCMC analyses should include local distance-ladder constraints such as SH0ES and compare the resulting V9 chains directly against matched Λ CDM chains using the same likelihood combination. This test is especially important because the present V9 run already shifts H_0 upward without using a SH0ES prior.

Fourth, additional late-time and structure-growth datasets should be included. These include CMB lensing, ACT/SPT high- ℓ CMB data, weak-lensing surveys, and large-scale-structure likelihoods. Such tests are necessary to determine whether the simultaneous increase of H_0 and reduction of S_8 remains stable under extended data combinations.

Finally, the present paper does not address primordial inflation. The current implementation assumes post-inflationary initial conditions and studies the subsequent effective vacuum evolution. A primordial extension of the same inventory principle may be investigated separately, but it is not part of the present analysis.

11 Conclusions

We have introduced an inventory-driven vacuum cosmology in which dark-energy evolution emerges from the growth of closure inventory.

Preliminary MCMC analyses indicate that the framework remains competitive with Λ CDM while shifting the inferred Hubble constant toward larger values.

A particularly attractive aspect of the model is that the effective equation of state $w(a)$ emerges dynamically from the inventory evolution rather than being imposed *a priori*. Although preliminary, the present results suggest that the closure-inventory principle may provide a promising phenomenological framework for post-inflationary cosmology and deserves further investigation.

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